Quadratics: Expanding & Factorising

Practice test 5 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 6)(x + 8)

[2 marks]

A2. Expand and simplify (x – 10)(x + 4)

[2 marks]

A3. Expand and simplify $(x + 9)^2$

[2 marks]

A4. Expand and simplify (5x – 2)(2x + 3)

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (x + 4)(x – 3)(x + 1)

[3 marks]

B2. Expand and simplify $(2x + 3)(x – 1)(x + 5)$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 11x + 24$

[2 marks]

C2. Factorise $x^2 – 8x – 33$

[2 marks]

C3. Factorise $x^2 – 121$

[1 mark]

C4. Factorise fully $3x^2 – 27x + 60$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $2x^2 + 11x + 12$

[2 marks]

D2. Factorise $3x^2 – 7x – 6$

[2 marks]

D3. Factorise fully $6x^2 - 3x - 30$

[2 marks]

Section E — Using factorising

E1. Solve $x^2 + 3x - 40 = 0$

[3 marks]

E2. Simplify fully

$$\frac{x^2 - 25}{x^2 + 2x - 15}$$

[3 marks]

Section F — Problem solving & proof

F1. (a) Write $x^2 - 10x + 18$ in the form $(x + a)^2 + b$. (b) Hence write down the coordinates of the turning point of $y = x^2 - 10x + 18$.

[3 marks]

F2. n is an integer. Prove that $(n + 3)^2 - (n + 1)^2$ is always a multiple of 4.

[3 marks]

F3. Without a calculator, work out $65^2 - 35^2$

[1 mark]